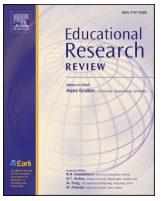
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Review



Research on research use in schools: A systematic review with a framework synthesis

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ABSTRACT

This systematic review investigates how use of research in schools has been framed and described in empirical literature, identifying the prevalent barriers and enablers reported. Systematic searches were conducted in January 2025 across Scopus, Web of Science, and ERIC databases, yielding 23,299 records after duplicates were removed. After a double-blind screening process in two stages, 109 studies were included based on a priori eligibility criteria: studies focusing on engagement with research among teachers and school leaders in primary and secondary education, published in English-language, and in peer-reviewed journals. Study quality was appraised using the Mixed Methods Appraisal Tool (MMAT) through a double-blind process. Data were synthesized based on an initial framework consolidating recent conceptualizations and theory on research use in education The findings highlight that while research use is a complex phenomenon shaped by systemic influences, it has been studied primarily at the individual level. The analysis identified four governing mechanisms - trust, relationships, skills development, and sustainable structures - that span from outside the system through the organizational level to individual educator practice. Consequently, in this review we propose a conceptual shift from "implementation" of research findings toward an "integration" of research with other knowledge sources and professional wisdom through thoughtful engagement. This interpretation suggests that moving beyond symbolic appropriation of research requires fostering professional agency through sustainable structures, rather than relying on instrumental mandates of evidence-based practice.

1. Introduction

Even though the use of research evidence (URE) to inform and improve practice has been a stated goal for education systems across the globe for two decades, the results of multiple international tests and surveys on student outcomes show a decline in many areas. This occurs despite a steady flow of reforms in many participating countries over the same period (OECD, 2023a; Perry & Morris, 2023; Sahlberg, 2023). One influential idea that has spread across school systems since the millennium is that of evidence-based practices (Røvik & Pettersen, 2014). Although aimed at providing direction for targeted practices, the effort to standardize teaching in an era of data-driven accountability may have led to a symbolic use of research evidence resulting in a shallow layering of new practices over traditional ways of thinking. While student outcome declines are due to myriad factors, the prevailing emphasis on standardized evidence-based practices may have failed to move systems in the desired direction. Achieving meaningful impact may instead be more

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closely associated with thoughtful use of research, than with its symbolic appropriation or the instrumental application of pre-packaged evidence, which can vary significantly in quality.

The global focus for research to inform policy and practice within education as well as in other sectors (e.g., Farley-Ripple et al., 2022; OECD, 2021, 2022; Rickinson et al., 2023) has also resulted in a growth in interest to study URE. Several systematic reviews (Dagenais et al., 2012; Dyssegaard et al., 2017; Hemsley-Brown & Sharp, 2003; Rickinson et al., 2022), reports (Dyssegaard et al., 2017; Finnigan, 2021; Hillage et al., 1998; OECD, 2022, 2023) and research projects (Farley-Ripple et al., 2022; Mausethagen et al., 2023; Rickinson et al., 2023) have contributed to a comprehensive knowledge base in the field, and a growing body of theory. However, they address a rapidly expanding field, and there is still a need to further conceptualize research use in schools, and to go deeper into the mechanisms that govern such use. The aim of this systematic review is to get an updated overview on how research use in schools is framed and described in empirical research, and what barriers and enablers of such use are described in the existing research literature. Specifically, it aims to investigate how these empirical findings align with a consolidated multi-level framework of research use. To incorporate the existing knowledge base, this systematic review employs framework synthesis to map and interpret the emerging themes in the literature.

1.1. Research use in education

A wide range of concepts and terms, such as evidence-based practice, evidence-informed practice, and knowledge mobilization, has been used to describe the phenomenon of URE, each carrying its own definitions and configurations, with variations that are often subtle. The broader literature also includes theories and studies of dissemination and implementation. In this systematic review, the term "use of research (evidence)" (URE) is used as an umbrella term to cover all forms of research use and was referred to throughout as "research use", "use of research", or URE. The definitions used in this review are drawn from Finnigan (2021), where research evidence (RE) refers to systematic empirical studies employing rigorous processes, including qualitative, quantitative, or mixed-methods designs. URE involves actively engaging with such research evidence to inform decision-making or practice (Finnigan, 2021).

Discussions on the utility of educational research have been present since educational science emerged in the beginning of the 20th century (Kvernbekk, 2016; Perry & Morris, 2023). The discourse gained new traction following professor of sociology David Hargreaves' address to the Teacher Training Agency in London in 1996, and through the ensuing debate with fellow sociologist Martyn Hammersley (Hammersley, 1997, 2000; Hargreaves, 1996, 1997). This debate centered around whether teaching and educational research could become research-based professions, utilizing experimental research methods and building a cumulative knowledge base similar to that in medical research and practice.

In the wake of this debate, the "What works" movement emerged as a prominent paradigm in education policy and research following Robert Slavin's (2002) proposed architecture of using randomized controlled trials to test and "prove" the effect of pedagogical practices and programs. Subsequently, large-scale meta-analyses, such as John Hattie's (2008) "Visible Learning", popularized evidence-based practice and introduced concepts like "effect sizes" into the educators' professional vocabulary. At the same time, clearinghouses such as the WhatWorks Clearinghouse in the US and the Education Endowment Foundation in the UK played a central role in this movement by evaluating the methodological quality of research and acting as brokers of the evidence. However, the "What Works" movement's assumption that the identification and dissemination of effective interventions would in itself lead to meaningful changes in practice was not as effective as expected. This further contributed to the emergence of implementation research (e.g., Cook & Odom, 2013; Fixsen et al., 2005), which emphasized translating evidence into educational practice.

This shift towards prioritizing research use has also been reflected in educational reforms worldwide, which have increasingly focused on integrating research findings into educational policy and practice. In response to, or perhaps serving as a catalyst for, this development, the OECD's Centre for Educational Research and Innovation (CERI) has been an influential contributor. In 2021, CERI launched the project "Strengthening the impact of education research" to explore the barriers and enablers of research impact, how to assess its quality, and how to engage different actors in its production and use.

Through a policy survey and collection of promising practices, the project produced two reports (OECD, 2022; 2023). The first report called for a conceptual shift from brokerage to engagement, arguing that policy and practice considerations must be connected at an abstract level. It also warned against top-down checklists, advocating instead for more meta-research to determine "what works in what works" (Révai, 2022a, p. 243). The second report concluded that developing a culture of research engagement required various stakeholders to enable thoughtful engagement. Since direct and instrumental use of evidence is seldom possible in complex systems, thoughtful engagement demands the motivation to challenge one's views based on research. Such a culture of exploration and creativity can only thrive in a trusting environment (Révai & Mouthaan, 2023, p. 2).

During the same time period, there has also been an expansion of mechanisms that do not support a trusting environment, and thus research use in education (Godfrey, 2019; Sahlberg, 2023). The driving forces of new public management principles, such as increasing competition between schools, standardization of teaching and learning, priority on basic literacy and numeracy, test-based accountability and increased privatization, do not encourage trust and innovation (Godfrey, 2019; Sahlberg, 2023).

Sahlberg (2023) characterizes the application of free-market principles in education policy as part of the Global Education Reform Movement (GERM) and draws on the global response to the 2020 pandemic to propose alternative policy directions that may provide professional agency to integrate research findings rather than standardize practice. He argues that

> *Success in fighting the global pandemic is a result of relying on best available science and expert knowledge to maximize the effectiveness of treatments while minimizing their side effects. We should follow that same principle that have led success in public health in many parts of the world in education, too.* (Sahlberg, 2023, p. 7)

While Sahlberg (2023) emphasizes how systemic policy mechanisms can restrict professional agency, research consistently shows that practical and cultural barriers within schools remain equally persistent. Despite decades of discussion, both the explanations for this limited engagement and the proposed solutions have remained largely consistent over the past 26 years (Mausethagen et al., 2025; Mausethagen & Hermansen, 2023). Commonly cited barriers include a lack of research knowledge, absence of effective brokerage, and unsupportive professional cultures (e.g. Hemsley-Brown & Sharp, 2003), along with practical challenges such as insufficient time, limited resources, and restricted access to academic literature and networks (e.g., Dagenais et al., 2012). Yet, these recurring lists of barriers offer little insight into what research engagement looks like when it does occur (Mausethagen et al., 2025; Mausethagen & Hermansen, 2023). In response, recent studies have both called for, and in some cases shifted focus toward, exploring how teachers and school leaders *actually* engage with research in practice.

1.2. Previous systematic reviews on URE and research gaps

Over the past two decades, several systematic reviews have examined how educators engage with research and what factors enable or hinder this process. While the reviews provide valuable insights, they also reveal important limitations that inform the need for further studies.

The earliest of these, by Hemsley-Brown and Sharp (2003), compared research use in education and medicine. Their analysis showed that although research could support professional development and organizational change, its application in schools was often constrained by cultural and institutional norms. They emphasized the role of leadership and organizational culture in shaping attitudes toward research, alongside structural factors such as legislation, funding, and dissemination strategies. A central recommendation was that policy efforts needed to be strengthened to bridge the gap between academic knowledge and classroom practice.

A different perspective was offered by Dagenais et al. (2012), who focused exclusively on schools. Their review highlighted methodological weaknesses in the existing literature: small sample sizes and inconsistent designs, which limited conclusions about what drives research use. Despite these limitations, they reinforced the importance of collaboration between researchers and practitioners, while adding that policy connections were critical for sustaining research-informed practice.

Further narrowing the lens, Dyssegaard et al. (2017) examined primary and lower secondary education. Their findings underscored the pivotal role of school principals and instructional leadership in implementing research-based programs. They also pointed to the importance of sustained professional development, peer collaboration, and trust as conditions for success. At the same time, barriers such as insufficient support, unrealistic expectations, and high staff turnover were identified as persistent challenges.

Finnigan (2021) approached the topic from a different angle by synthesizing gaps in the knowledge base. The methodological gaps underscored the needs for a) conceptualizing quality use, b) developing measurement tools of URE, c) employing longitudinal studies, and d) conducting qualitative and mixed-methods studies to further develop or revise conceptual frameworks. This work also called for targeted efforts to strengthen researchers' capacity to transfer evidence into practice and highlighted the political dynamics that shape research adoption. Finnigan (2021) warned that without such efforts, evidence risks being misinterpreted or misused in policy and practice.

An answer to the call for conceptualizing quality use was allready underway when Rickinson et al. (2022) introduced the concept of "Quality Use of Research Evidence" (QURE) as a result of a systematic review mapping quality use of research across four sectors. The framework defines quality use as thoughtful engagement and implementation of appropriate research, enabled by several individual, organizational, and systemic factors. While promising, the framework remains in its early stages and will require empirical testing to refine its applicability.

Most recently, Mausethagen et al. (2025) conducted a systematic critical review of 34 articles examining how research use has been conceptualized over time in education. Their analysis identified a persistent deficit framing in the literature, namely that research use is consistently presented as something that is lacking or insufficient. They also argued for more profession-sensitive conceptualizations that take seriously the complexity of teachers' epistemic work. While their review focused on conceptualization and identified the need for frameworks that better capture professional agency, it did not undertake an empirical synthesis of barriers and enablers or propose a theoretical framework.

Together these reviews have addressed complementary dimensions of the field: recurring barriers to research use, the importance of leadership and collaboration, and the need for stronger links between research, policy, and practice, while differing in focus, scope, and type of synthesis. However, there is a further need to develop or revise conceptual frameworks and address the mechanisms that govern the persistent barriers found, or as put by Révai (2022, p. 241): "finding out what works in what works". While former reviews identified the 'what' of research use, they have still left a gap regarding how broader societal and systemic developments govern the internal capacity of schools to move beyond symbolic and standardized instrumental use. This review addresses this gap by proposing and testing a consolidated framework against recent empirical data. The initial consolidated framework is based upon several conceptual frameworks and theoretical constructs.

1.3. Conceptual frameworks on URE

To deepen understanding of how research use unfolds in practice, several conceptual frameworks have also been developed in recent years. These frameworks conceptualize research utilization in various forms, and some propose mechanisms that enable or hinder engagement and offer lenses for interpreting the complexity of research use in schools.

An early example of conceptualization, and terms that have been used throughout this paper, comes from Weiss (1979) who introduced seven models of research use (i.e., the knowledge-driven model, the problem-solving model, the interactive model, the

political model, the tactical model, the enlightenment model, and research as part of the intellectual enterprise of society). This served as the basis of the later typologies of research use as conceptual, instrumental or symbolic (e.g., Knorr, 1977; Nutley et al., 2007; Pelz, 1978). Conceptual use comprises a wide-ranging definition of research use, where research informs practitioners in complex and often indirect ways, and has an impact on the understanding and thinking of various issues (Nutley et al., 2007), or as Weiss (1980, p. 381) puts it; "knowledge creep".

Instrumental use, however, refers to the direct impact of research on policy or practice decisions, where a certain piece of research directly addresses a certain issue (Nutley et al., 2007). Symbolic (or strategic/tactical) use, on the other hand, refers to a way of using research to predominantly post-hoc legitimize already made decisions (Knorr, 1977).

A later influential contribution, also from the US, partially built on these concepts. Farley-Ripple et al. (2018, 2022) introduced the "deep use" framework, which concerns what kinds of evidence are used and also how they were used, whether instrumentally, strategically, or conceptually, and how educators search for, interpret, and integrate research into their work. A large mixed-case study applying this framework revealed substantial variation across schools, even among those classified as "deep users." (Farley-Ripple et al., 2022). What united these schools, however, was the integration of research use into their culture, structures, and routines. Leadership, capacity, relationships, and brokerage emerged as enabling conditions, yet challenges persisted, including differing notions of what counted as research and difficulties in accessing and implementing relevant evidence.

A Norwegian case study by Mausethagen and Hermansen (2023) addressed similar challenges but through a different lens: research literacy. Drawing on Eriksen's (2022) conceptualization, the study sought to showcase the activities and behaviors of research use, particularly within the dichotomy of "teachers as researchers" and "teachers as recipients of research." Rather than seeing professional practice through an art/science dichotomy, Eriksen (2022) argues for viewing profession through a craft model. This model features integrity, reason-giving, and flexibility as the main attributes of professionality. Within this framework, research literacy is a virtue that preserves the integrity of the domain of expertise. It also serves a mediating function between the domain of "what works" and the domain of "what is appropriate." Research literacy and its three attributes are served by three sensitivities that apply when professionals engage with research. Genre sensitivity serves integrity by viewing practice as a complex genre. Practice sensitivity serves reason-giving by focusing on the intersubjective and audience-dependent justification of actions and choices. Finally, situational sensitivity serves flexibility by judging the appropriateness of an action or choice in a specific time, space, and situation.

In the case study by Mausethagen and Hermansen (2023), these sensitivities surfaced in how school leaders selected and adapted research, how teachers negotiated adherence and resistance, and how emerging challenges prompted demands for new knowledge. This perspective highlights research use as a process of re-contextualization: transformative, evaluative, and exploratory.

From Australia, the Monash Q-project introduced the QURE framework (Rickinson et al., 2022, p.142), which defines quality use as "thoughtful engagement with and implementation of appropriate research evidence, supported by a blend of individual and organizational enabling components within a complex system." Building on a systematic review and extensive empirical work, the framework identifies enabling factors at multiple levels: individual (skills, mindsets, relationships), organizational (structures, culture, leadership), and systemic influences that shape these interactions.

System-level dynamics are even more pronounced in Godfrey & Brown's (2019) ecosystem model for research-engaged schools. Synthesizing empirical studies and theoretical perspectives, this framework maps conditions across micro-, meso-, exo-, macro-, and chrono-systems, illustrating how research engagement is embedded within broader educational ecologies.

Finally, the conceptual foundation for the current review draws on these frameworks alongside ideas of internal and external complexity (Perry & Morris, 2023). Kvernbekk (2016) articulates these complexities by posing questions practitioners should consider when engaging with research - especially experimental studies. These include examining factors of the internal system in which research was produced and the external system where it will be applied, as well as considering side effects and alternative means to achieve desired outcomes. Such considerations are rarely addressed in educational research but are integral for practical use.

While these frameworks provide a rich account of URE, they currently operate at different levels of analysis with limited conceptual interface. For example, while Eriksen (2022) provides a deep philosophical account of teacher professional virtues, it does not explicitly integrate with the macro-level systemic pressures described in Godfrey & Brown (2019) ecosystem model. The different levels addressed, the different geographical contexts and the different constructs used in these frameworks do not necessarily mean that there are any tensions or contradictions among them, but the insights they offer remain somewhat fragmented. The current review, hence, aims to consolidate these insights and examine how they align with empirical evidence in the field.

1.4. The current systematic review

Building on the limitations identified in the previous reviews and the conceptual advances offered by the recent frameworks, this systematic review aims to provide an updated synthesis of empirical research and integrate theoretical perspectives into a coherent understanding of the mechanisms that govern URE.

The former reviews and reports have largely built up a cumulative knowledge base for URE. Hemsley-Brown and Sharp (2003b) reviewed the literature in the wake of the Hargreaves (1996, 1997) and Hammersley (1997, 2000) debate and concluded that cultural shifts in education to support URE need to start with policy. Policy changes and strategies in the early 2000s address this (Révai, 2022), and broker agencies, clearinghouses and WhatWorks-centres have promoted evidence-based or evidence-informed practice and policy. However, in the early 2010s, URE was still scarce in education, and Dagenais et al. (2012b) called for clearer conceptualization and more empirical research on URE. This corroborates Finnigan's (2021) recommendations for further research in the field. Answers to these calls came in the before mentioned projects in the US, Australia, the UK and Norway (Farley-Ripple et al., 2022; Godfrey & Brown, 2019; Mausethagen et al., 2023; Rickinson et al., 2023). These recent frameworks, however, differ in scope, ranging from

concepts of professional virtues and specific activities to broad systemic conditions.

The consolidated framework presented in Fig. 1 represents a multi-layered and integrative synthesis of current research on URE. The integration is achieved by mapping specific factors of research use on to a socio-ecological structure, moving from the individual practitioner to the broader educational system. At the center of the model, we integrate concepts from Rickinson et al. (2022), Eriksen (2022), Kvernbekk (2016), Perry and Morris (2023) and Farley-Ripple et al. (2018) to define both what is being used and how. The vertical columns of the table and the concentric circles of the diagram represent the ecological layers found in both Rickinson et al.'s (2022) and Godfrey & Brown's (2019) models. Key conceptual elements from these models are also incorporated in the table. The consolidated framework's purpose is to illustrate and conceptualize the multifaceted complexity of research use in education.

In this framework, and throughout the review, individual level refers to the micro system, the individual actors such as teachers and school leaders. The organizational level refers to the school, or in some cases where schools cooperate in a meso-level or local administrative system such as a municipality. System level refers to the macro and exo-system in the socio-ecological sense, encompassing national policy, educational governance structures, and broader societal factors, rather than any specific administrative configuration.

By consolidating these perspectives, this review attempts to capture the multi-level nature of URE and address the fragmentation noted in previous work. This systematic review, therefore, aims to contribute to the existing knowledge base by providing an updated review of empirical research and by offering a consolidated and coherent examination of the frameworks developed to understand and support URE.

To get a thorough and updated overview of various conceptualizations and components used in international research on URE, this systematic review addresses the following research questions:

- How is use of research evidence framed and described in empirical research on use of research in schools?
  - o What components of research use emerge in empirical research?
  - o What relations among these components are identified?
- Which barriers and enablers of research use in schools are described in extant empirical research?

To address these questions and incorporate existing theoretical frameworks, the study utilizes framework synthesis. This method is well suited to bring together empirical literature and theoretical perspectives to form an understanding of the issue under study into an initial or revised framework that evolves iteratively as new data are incorporated (Brunton et al., 2020).

Consolidated Framework for Research Use

Core	Individual level[1] Teachers and school leaders	Organizational level[1] Schools (and municipalities/districts)	System level[1] (the national/state/district education system)
Appropriate research[1]	**Skillsets**[1]	**Infrastructure**[1]	Policy
Genre sensitive[2]	Research informed professionals[3]	Support systems[3]	School institutions integral to wide societal aims and ambitions[3]
Practice sensitive[2]	Initial education and professional learning[3]	KM-networks[3]	
Situation sensitive[2]			
Deep use[4]**:**	**Mindsets**[1]	**Culture**[1]	Intelligent accountability[3]
Search	Attitudes and perceptions	Professional development[3]	Trust[3]
Interpretation	Trust	Learning communities[3]	
Participation		Adaptive[3]	
Frequency		Semi-permeable[3]	
Thoughtful engagement and implementation[1]**:**	**Relationships**[1]	**Leadership**[1]	Societal influences (Zeitgeist)
Governing and enabling factors of internal and external context[5]	(Mesosystem) Coherent professional bodies[3]	Intelligent accountability[3]	Belief in key role for practitioners and researchers[3]
System stability[6]	Universities and research organizations[3]		Strong inclusive narrative about importance of research informed practice[3]
Permissibility[6]			
Side effects[6]			

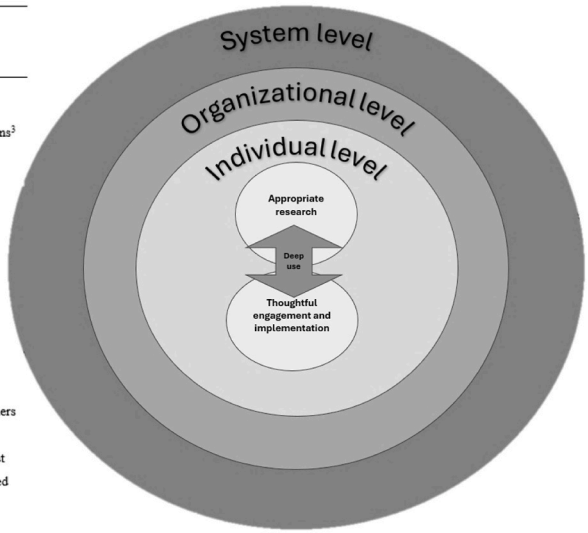


Fig. 1. Consolidated Framework for Research Use

Note. The initial framework for the framework synthesis, consolidating concepts from Rickinson et al. (2022) [2]; Eriksen (2022) [3]; Godfrey (2019) [4]; Farley-Ripple et al. (2018) [5]; and Kvernbekk (2016) [6] and Perry & Morris (2023) [6].

2. Method

2.1. Transparency and openness

To ensure transparency and openness, a protocol for this systematic review was pre-registered in the Open Science Framework (masked for the review process) before the formal literature search. There were no significant deviations from the protocol. Search strings used in various databases, extracted data and coding can also be accessed in OSF (will be made available upon acceptance). When reporting on this systematic review, we followed guidelines in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement (Page et al., 2021).

The review team consisted of three female researchers with diverse backgrounds from three nations, respectively, Norway, Finland and Türkiye. The first two authors have a background as practicing teachers, whereas the third author is a researcher with a background in psychology. The second and third authors have expertise in conducting systematic reviews, which was the method employed in this study. Our collective background and diverse competence allowed us to capture the complex nature of the topic. Although our positionality is shaped by our European context, this study draws upon a wide range of research, predominantly from the UK and USA. This weighting reflects the greater number of studies on the topic published in these countries compared to other countries, rather than an intentional Anglo-centric focus. By acknowledging this, we aim to clarify that our goal is to synthesize insights from existing research to inform a global audience.

The analytical approach in this review was informed by a critical realist perspective, which seeks to identify underlying mechanisms that generate observable patterns rather than merely describing surface phenomena (Bhaskar, 2013). This orientation shaped both the framework synthesis method and the configurative analysis, directing attention toward the conditions and mechanisms that govern research use across levels rather than toward frequency counts of barriers and enablers alone. Our theoretical orientation toward professional agency and socio-ecological frameworks, rather than, for instance, psychological or purely organizational models, inevitably sensitized the analysis to systemic and relational patterns in the data, and may have rendered individual-level factors relatively less prominent in the configurative synthesis than a differently oriented team might have found.

2.2. Literature search

Pilot searches were conducted during the fall of 2024 and involved an iterative process of developing keywords and search terms. As the focus of the review is “use” and “research”, which are very frequent words used in most research, the process of creating a search string that would both be sensitive enough to ensure a systematic result, and specific enough to yield a feasible workload, proved a challenge. After several iterations and consultations with other researchers in the field, the formal systematic literature search was conducted in January 2025. The terms used represented different variations in the concept of URE and terms indicating that the URE should take place in education, by teachers and/or school leaders. Example search string from SCOPUS database is shown below:

> (TITLE-ABS-KEY ("evidence-based practice*" OR "knowledge dissemination*" OR "knowledge transfer*" OR "knowledge mobilization" OR "knowledge mobilisation" OR "use of research evidence" OR "knowledge exchange" OR "information utilization" OR "information utilisation" OR "information dissemination" OR "information transfer" OR "knowledge communication" OR "research engaged" OR "research informed" OR "research in practice" OR "evidence-based teaching" OR "evidence-informed teaching" OR "research-based teaching" OR "research-informed teaching" OR "knowledge acquisition" OR "research use" OR "knowledge use" OR "research utilization" OR "research utilisation" OR "evidence use" OR "evidence informed" OR "use of evidence" OR "research engagement" OR "research literacy" OR "research implementation" OR "implementation of research" OR "knowledge-mobilization" OR "research-practice gap" OR "knowledge broker" OR "knowledge brokerage" OR "implementing research" OR "implementing evidence" OR "implementing knowledge" OR "implementation of evidence" OR "implementation of knowledge" OR "research-informed" OR "research-influenced" OR "research knowledge" OR "research engagement" OR "Research culture") AND TITLE-ABS-KEY (educator* OR school* OR teacher* OR principal* OR teaching OR leader*)) AND (EXCLUDE (SUBJAREA , "MEDI") OR EXCLUDE (SUBJAREA , "NURS") OR EXCLUDE (SUBJAREA , "DENT") OR EXCLUDE (SUBJAREA , "VETE")) AND (LIMIT-TO (DOCTYPE , "ar")) AND (LIMIT-TO (LANGUAGE , "English")) AND (LIMIT-TO (SRCTYPE , "j"))

The search was conducted in the databases of Scopus, Web of Science, and ERIC, and these databases were selected due to their relevance for educational research and their broad scope. For a comprehensive overview of the search string used in each database, see Appendix A.

2.3. Study identification

To ensure high quality, transparency and rigor in the screening process of this systematic review, eligibility criteria were set a priori, based on the overall aim and the research questions during the preparation stage of the review (see Table 1). The criterion on topic largely draws on the definition of URE from Finnigan (2021): *‘Use’ in URE refers to the process of actively engaging with and drawing on research evidence to inform, change and improve decision making or practice.* Other criteria described the target population, language, study type, and publication type, set to address the aim of the review. Early childhood education and care (ECEC) settings were excluded from the target population as these fall outside mandatory schooling in most national systems. The professional cultures, governance structures and accountability frameworks that shape research use in voluntary educational settings such as ECEC, differ

from those in compulsory primary and secondary education and would warrant separate investigation.

To ensure a broad international scope and capture the evolution of the field, no restrictions were placed on geographical location or publication year. The inclusion criteria were, however, limited to empirical studies published in peer-reviewed journals to ensure that the review was based on documented evidence of practice and to maintain a consistent standard of quality and search reproducibility. Recently developed theoretical and conceptual perspectives have been incorporated through the initial consolidated framework, which serves as the analytical lens for empirical data (Table 1).

To enhance inter-rater reliability and minimize bias, study eligibility was determined via a two-stage, independent screening process using EPPI-Reviewer software (http://eppi.ioe.ac.uk/cms/). The screening process comprised an initial review of titles and abstracts, followed by a full-text assessment. At each stage, a pair of reviewers conducted independent, blind screenings. When reconciling the two screeners' work, disagreements were resolved through discussions, and with the involvement of a third reviewer. The review team met regularly to facilitate reconciliation and discussions.

2.4. Data extraction and synthesis: framework synthesis

The current review uses framework synthesis, partly because it gives a structured, yet flexible approach to handling diverse empirical data of all designs – but also because the method is well suited to integrate existing theories and structure data analysis around pre-defined themes and allowing for new themes to emerge (Brunton et al., 2020). Framework synthesis stems from framework analysis, which was originally used to analyze research data to inform policy concerns. The framework develops iteratively, as the research questions, the theoretical background and the empirical literature reviewed shape an understanding of the issue at hand (Booth et al., 2022; Brunton et al., 2020). This allows for an initial conceptual framework to evolve during the synthesis (Gough et al., 2017). In this review, the "initial conceptual framework" refers to the a priori framework that was developed in the preparation stage for the review. With the configurative synthesis of the results, the framework can be revised if necessary. The review followed the stages for framework synthesis as outlined by Gale et al. (2013). These guidelines outline the methodological steps in the process, beginning with the formulation of the problem and the framing of research questions, followed by the search for and identification of eligible literature. Subsequently, data were extracted, synthesized, and analyzed, and study quality was assessed before integrating and presenting the findings. This involved an initial literature review, to get familiarized with the field of URE and identify theoretical frameworks that had been developed recently to explain and conceptualize URE. The main framework stems from the QURE framework (Rickinson et al., 2022), but with themes from other frameworks placed within it as sub-themes or explanatory concepts.

In the core, the deep use framework (Farley-Ripple et al., 2018) outlines activities, and in which manner and circumstances they take place in the arrow between "thoughtful engagement and implementation" and "appropriate research". To elaborate on "thoughtful engagement", concepts from Kvernbekk (2016) and Perry and Morris (2023) were used. "Appropriate research" is guided by the sensitivities to research literacy as suggested by Eriksen (2022), even though this framework has a much broader remit. The individual, organizational and system level of the framework, all have subthemes incorporated from eco system for research engaged schools (Godfrey & Brown, 2019). In addition, we incorporated two sub-themes to the system level that we believe exert an impact on URE, namely general societal factors such as public discourse and trends, and educational policy and politics.

In this framework synthesis, data extraction and coding were carried out in two phases. In the first phase, to gain a detailed overview of the existing research, descriptive information on each study was extracted in an excel file. The information extracted in this phase included characteristics such as publication year, country of study origin, study and data design, data collection methods, study informants and study subjects, educational level of the study object and main theme of the study.

For the second phase of data extraction, the included studies were coded in NVivo 15, using the main concepts from the research questions as primary nodes: framing and descriptions, barriers, and enablers. Data were extracted from results sections, and in some cases the discussion, of included studies. The extracted data in the "barriers"-node were used for pilot coding and inductively coded by the members of the review team in a blind process to enhance inter rater reliability. The pilot extraction showed a high agreement between coders, and discrepancies were resolved through consensus-based discussion until 100% agreement was reached. The

Table 1
Eligibility criteria.

Criteria	Inclusion	Exclusion
Topic	The study must contain descriptions of (population) actively engaging with and drawing on research evidence to inform, change and improve decision making or practice (i.e. searching, interpreting, deliberating, discussing, trying out, reflecting on, action research based on RE, response to programs or evidence practices).	The study does not contain descriptions of engagement or use. This excludes school wide implementation of research-based programs or evidence-based practices -unless the study describes teachers/school leaders actively engaging with and drawing on research evidence *in* the intervention/program/EBP.
Target population	Teachers or school leaders working in primary or secondary education	District level school leaders, school board members, teachers/leaders in higher (tertiary) education, teachers/leaders in ECEC, pre-service teachers, researchers, students, school librarians/nurses/social workers/ psychologists
Language	English	Not English
Study type	Empirical studies employing qualitative, quantitative or mixed methods designs	Theoretical or conceptual papers, commentaries, secondary studies
Publication type	Peer reviewed journals	Book chapters, grey literature, reports

enablers node and the framing and descriptions node were coded by the first author alone, with selected extracts discussed and verified in the review team to ensure consistency of interpretation. This process resembles a thematic synthesis, but as the initial framework is in place beforehand, the coding is not as open ended as a thematic synthesis at the outset (Gough et al., 2017).

The analysis then proceeded in two stages. First, the codes from the QURE-framework were analyzed aggregately by summing up and identifying the number of studies from which they originated. While the aggregative analysis mapped findings onto the QURE framework, the subsequent configurative pass identified themes that crossed framework levels. In this second phase we went deeper into coded data in each category and identified themes across different categories. As an example, the coded data for "relationships" carried different meaning on the individual, organizational and system level. Both the barriers/enablers and framing/descriptions data were discussed iteratively in the review team until consensus was reached on four governing mechanisms. This process involved writing out analytical texts for each suggested mechanism, exemplified with representative quotes from the included studies, and holding analytical discussions in the review team based on the extracted data.

The identified themes were not pre-defined in the initial framework but surfaced through the configurative stage of the analysis, in which the review team examined the aggregated codes across framework levels and identified patterns that could not be contained within a single level. Specifically, themes that appeared repeatedly across at least two levels, and in both the framing/descriptions and barriers/enablers data, were brought forward for iterative team discussion. Through this process, consensus was reached on four themes that function not as level-specific components but as potential governing mechanisms that shape research use across the entire system: trust, skills development, sustainable structures, and relationships.

It is worth mentioning that while another team working with the same data might have weighed or labeled these mechanisms differently, the consistency with which they appear across levels, study designs, and national contexts lends confidence to their identification within our team.

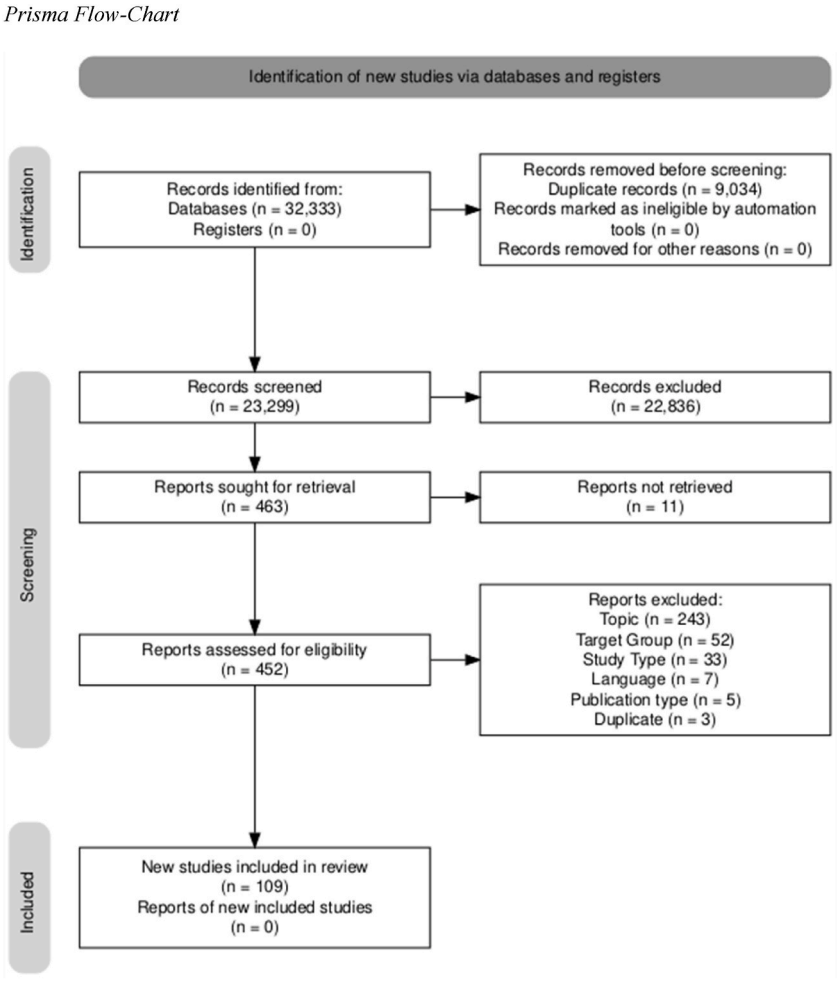


Fig. 2. Prisma Flow-Chart
Note (Haddaway et al., 2022).

2.5. Study quality

To critically appraise the included studies, we used the Mixed Methods Appraisal Tool (MMAT; Hong et al., 2018). The tool has five specific criteria for each type of research design, encompassing qualitative, quantitative, and mixed methods studies, in addition to two universal screening questions to determine whether the quality can be appraised. A pair of reviewers appraised each of the included studies, in a double-blind process, before comparing and resolving discrepancies with a third reviewer. Findings from lower-quality studies were only reported where corroborated by at least one higher-quality study, thereby ensuring that themes included in the synthesis were primarily grounded in methodologically robust evidence. Where a low-quality study was the sole source for a finding, this was explicitly noted in the analysis and interpreted with caution. MMAT ratings were also used during the interpretive phase of the synthesis to assess the strength and consistency of evidence across studies.

In a framework synthesis, the appraisal process may also require assessment of the theory or framework that has been identified for analysis (Booth & Carroll, 2015). No specific tool was used for this, but all included theories and frameworks come from peer reviewed literature, have conceptual clarity and operationalizability, and are fit for our data.

3. Results

3.1. Study identification

A total of 32,333 potential records were retrieved from searches in SCOPUS ($k = 12,812$), Web of Science ($k = 10,420$) and ERIC ($k = 9101$), and uploaded to EPPI-Reviewer. After removing duplicates ($k = 9034$), 23,299 articles were independently screened based on title and abstract. With an overall agreement rate of 98,1%, 463 articles were found eligible for full text screening. Out of those, we were able to retrieve the full text for 452 studies. Eventually, 109 articles were included in the current systematic review, with an agreement rate of 81% for the full text screening (see Appendix B for the list of the included studies and Appendix C for their characteristics). Fig. 2 shows the details of the search and screening process.

The descriptive data show that the included studies stem from 28 different nations, but with the majority from English speaking countries (57%, UK = 27, Ireland = 1, USA = 15, Canada = 13, Australia = 5, New Zealand = 1). The remaining studies came from Europe ($k = 23$), Asia ($k = 14$), and Africa ($k = 2$). Eight studies involved multiple countries. The first eligible study was from 2004, but there has been a growing interest in the field over the past decade.

In terms of the study design, 38 studies (35%) employed quantitative designs (mainly utilizing surveys as data collection methods), 48 studies (45%) were qualitative (with mainly interviews as data collection methods) and 23 studies (20%) used multiple or mixed methods. Most of the studies (approximately 80%) were cross sectional.

3.2. Quality assessment

We used the MMAT appraisal tool (Hong et al., 2018) for quality assessment for different research designs. One study did not pass the screening and was not assessed. 41 studies (38%) were assessed to be of high quality, with a top score of 5. 18 studies (17%) were assigned a score of 4, and 31 studies (28%) were scored 3. The rest of the studies (17%) were assessed to be of low quality with scores of 2 or below. Appendix D shows the summary of the quality assessment for each study as well as aggregated scores according to research design.

Studies were not excluded due to quality, but the variation in quality was discussed, and considered when interpreting the results. The majority of findings reported in the synthesis were drawn from or supported by the 59 studies rated 4 or 5 on the MMAT, indicating that the main themes of the review were largely based on studies assessed as having high methodological quality. The findings appearing only in lower-quality studies were treated with greater caution in the discussion and are not used solely in drawing main conclusions.

3.3. Research question 1: framings and descriptions of URE in schools

The included studies, in general, show a multi-faceted and broad picture of research use in schools. They examine the extent to which research is used, how it is used, and some try out different interventions and methods to increase research use. What they all have in common, however, is that the lack of research engagement constitutes a deficit or a challenge in the profession, in line with the findings from Mausethagen et al. (2025). Using our framework as a lens to observe this through, we can see that the problems, and to some extent the solutions, can be framed at the individual level, the organizational level and/or the system level. However, our aggregative analysis showed a skewed emphasis where the individual level factors, and the core components of research engagement and sources of research, have had a greater focus than the organizational and system level factors (See Appendix E).

In the following sections, we present the results from our aggregative analysis through the different levels and based mainly on the components of the QURE framework, which for ease of analysis were used as the aggregated codes. The relations between components in the different frameworks that made up our initial framework are presented in the discussion section.

3.3.1. System level ($k = 46$)

Findings in the included studies showed that on the system level, the descriptions focus on the larger questions on how research, research use, and educators' and researchers' roles have systemic impact. In the studies, there seems to be a tension between two

opposing models of research use (e.g., Ahmed, 2019a; Farley-Ripple, 2024; Finnigan et al., 2013). The first is a policy driven, instrumental and simplistic expectation of basing educational practice on evidence. There is, however, no systemic incentive for research use, as the accountability systems measure student attainment, rather than collaborative practices in schools. Rather than raising expectations and policies to direct research use, systemwide investments in individual and organizational capacity building were suggested (Farley-Ripple, 2024).

The alternative model that is described in the included studies, on the other hand, idealizes the agency and professional responsibility of practitioners in schools as a driving force for research use. Here, research use is a form of collective sense-making. Effective research use is recast as a "broader social process" of "collective negotiation" (Hudson et al., 2016 p. 35). The system's goal should be to create structures that support this negotiation - providing space for "long, focused discussions" (Cain, 2015b p.482) where evidence is reinterpreted and given meaning within a specific context, rather than merely implemented. In the reviewed literature, the most prevalent framing is the first "top-down" model, whereas the second "bottom-up" model is seen as an ideal.

3.3.2. Organizational level ($k = 56$)

Based on our analysis of the organizational level, the concepts: culture, structure, and leadership are tightly knitted together. The *organizational culture* ($k = 41$) is described both as non-existing, where research engaged teachers feel "weird" or "in isolation" (Williams & Coles, 2007 p. 192-193) and as an embedded and intangible ethos of the school (Brown, 2017). In these cultures, research use is a collaborative endeavor, characterized by shared language, open dialogue, and debates (Baan et al., 2023; Gleeson et al., 2023; Mills et al., 2025). And as Ion and Sirvent (2022 p. 623) found, the role of trust is "fundamental". This is connected to a climate for innovation, and an environment where the risk of failure is highly accepted. As described by Gaussel et al. (2021, p. 9) "teachers thus need to know that someone "has their backs" should something go wrong".

The *organizational structure* ($k = 32$) is described in the reviewed literature as formal resources, integrated routines, and policy. The most critical formal resource is the allocation of time (Baildon & Ong, 2022; Gleeson et al., 2023; Lysenko et al., 2015; Mills et al., 2025), but there are also descriptions of resources in the form of different roles and responsibilities (Baan et al., 2023; Gleeson et al., 2023; Mills et al., 2025), and access to materials and other resources (Lysenko et al., 2015; Williams & Coles, 2007). The studies indicate that integrated routines like professional learning communities, or other collaborative forums, are means through which research use can move out of individual responsibilities and into a joint endeavor (Baildon & Ong, 2022; Brown & Zhang, 2017; Cain, 2015b).

But the central catalyst described for both components seems to be *leadership* ($k = 20$). In the included studies leadership is framed as an important component through descriptions of modelling and creating visions (Brown & Zhang, 2017; Gleeson et al., 2023; Mills et al., 2025), through empowerment and support (Baan et al., 2023; Farley-Ripple, 2024; Ion et al., 2024; Mills et al., 2025) and through strategic bridge-building (Cornelissen et al., 2017; Graves & Moore, 2018). In the reviewed literature, we found school leaders who model and embed research use in the school's culture (Brown & Zhang, 2017; Ion & Sirvent, 2022), and school leaders who use research symbolically to justify their own policy (Lysenko et al., 2015; Penuel et al., 2017; Wescott, 2022). There are also leaders who champion their teachers in research use, encouraging risk taking and innovation (Baan et al., 2023) – and there are leaders who actively hinder engaged teachers (Wescott, 2022). School leaders are both described as the most important knowledge brokers in the schools or their district (Finnigan et al., 2013), and as gatekeepers of both innovation (Wescott, 2022) and relationships with universities and research institutions (Prendergast & Rickinson, 2019).

3.3.3. Individual level ($k = 91$)

The descriptions on the individual level (present in 83% of the studies included) are characterized by a long series of gaps and sometimes contradictory themes. In the included studies, both teachers and school leaders reported both low levels of confidence (Farley-Ripple, 2024) and high levels of confidence in consuming and appraising research material, but this confidence is not always backed by actual skills in engaging with research (Ilhan et al., 2015; Mills et al., 2021). There are diverse and often contradictory descriptions of the *skillsets* ($k = 39$) of teachers and school leaders.

First, the skills to engage with research range from educators who are unaware of what constitutes educational research, through descriptions of different levels of research and data literacy, and to translational and application skills, where school leaders and teachers show the skills to adapt and translate research findings to their own context and practice. However, the latter might be framed differently in different studies. In some studies (e.g. (Malin et al., 2019; Riordan, 2022; Yoshizawa, 2022), the research use is shallow, symbolic, or instrumental to reach other means, and the research or evidence is diluted or conflated to fit the purpose.

On the other hand, there are examples of research use where "teachers were coming up with their own understanding and interpretation" (See et al., 2016 p. 11) or were practicing their own "personalization of evidence-based practices" (Guckert et al., 2016 p.74,76). Personalization, understanding and interpretation could also be seen as part of the process of adaptation or translation of research findings into practice, i.e., framed as a positive attribute to research use, so how it is framed might reflect the authors' view on both different kinds of use, and different kinds of research evidence.

In the included studies in this synthesis what seems to be a clear indicator of a good skillset for engaging in research regardless of framing is that the teacher or school leader has experience with research themselves, either through formal education, professional development courses or collaboration with researchers or action research (Farley-Ripple, 2024; Kowalczuk-Walędziak et al., 2024; Ståhle, 2022; * Thomm et al., 2021). The latter two also seemed to improve the confidence levels of practitioners when engaging with and in research, by demystifying the academic sphere (Bergmark, 2022).

The development of skillsets seems to be intrinsically linked to *mindset* ($k = 47$). Individuals' mindsets are described through their beliefs, attitudes and dispositions toward research and research use and framed through some key dimensions. The first is that there is a

value-action gap: teachers (and school leaders) value educational research, but do not use it (Baildon & Ong, 2022; Booher et al., 2020; Farley-Ripple, 2024; Gleeson et al., 2023; Konstantinou, 2022; Ovenden-Hope & la Velle, 2015; Penuel et al., 2017; Prendergast & Rickinson, 2019; Shi et al., 2022; Tunison, 2020a; Williams & Coles, 2007). The second dimension is that there is a spectrum of dispositions and attitudes towards research that goes from skepticism - the belief that research is disconnected from practice, to enthusiasm - which is often connected to a mindset where efforts for change are worthwhile (Cain, 2017; Déchène et al., 2024; Gleeson et al., 2023; Mockler & Stacey, 2021; Nägel et al., 2023; Vanderlinde & van Braak, 2010). Third, there is the aspect of trust and credibility (Ahmed, 2019b; Cooper et al., 2017; Déchène et al., 2024; Drill et al., 2013; Fraser et al., 2018; Gleeson et al., 2023, 2023; Graves & Moore, 2018; Hudson et al., 2016; Ion et al., 2019; Ion & Sirvent, 2022; Rosman & Merk, 2021; Rycroft-Smith & Stylianides, 2022; Schmidt et al., 2022). Teachers and school leaders put trust in familiar and credible sources, sometimes through their relationships. For teachers to trust research and find it credible, it needs to align with their beliefs, to be recommended by someone they trust or come from a known expert. However, their own professional ethics, and the shared values of the practice community, always seem to play a larger role than research. The research itself only gets to be a "third voice" (Cain, 2015b p.482), even when it is credible and trustworthy.

In an overlap position between individual mindsets and skillsets, the epistemological stance of teachers and school leaders, can shape how they perceive their professional endeavors – as an "art" where personal experience and ethical judgement is opposed to the instrumentality of research use (Ahmed, 2019a; Cain, 2015b), or a more scientific stance where professional ethics entail finding out and doing "what works" (Boyne & Beadle, 2017; Friesen & Brown, 2024).

In the included studies, *Relationships* ($k = 39$) within and outside the context of the school are framed as the essential social infrastructure for the individual use of research. These relationships are described as both informal and formal structures that exist on all levels of the ecosystem; from the colleagues sharing the same classrooms, through informal communities of practice and formal professional learning communities, to teacher unions, research institutions and intermediaries at the system level. Some studies have also found that formal structures for collaboration increase the informal social dimension of collegial interaction around research (Cornelissen et al., 2017). Many studies identify social media platforms as actual or possible personal learning communities for professionals, and two studies delve into how researchers and practitioners interact in these forums (Aukland, 2022; Déchène et al., 2024). An interesting observation from Aukland (2022) was that most of the interactions between practitioners revolve around practical tips and sharing of experiences in the classrooms, but that researchers – who themselves in this case are practicing teachers in teacher education, do not discuss their teaching practices.

3.3.4. Core components ($k = 94$)

The core components of the QURE-framework are "thoughtful engagement and implementation" and (of) "appropriate research", and their interconnectedness is represented by a two-way arrow. We have chosen to treat this arrow as a representation of meta-activities of research use, mainly because the data gathered that involved how research was sought and the frequency of these activities, do not necessarily constitute engagement, but are still components of use. "Appropriate research" ($k = 59$), as described in the included corpus, can be summed up under four themes. The first, and perhaps most prominent, one is *context*. The research needs to match or fit the school culture and local student demographics to be valued by practitioners (Ahmed, 2019a; Gleeson et al., 2023; Neal et al., 2019). The second theme is *practicality*. Within this theme are both accessibility – in terms of language, form and convenience (Gleeson et al., 2024; Konstantinou, 2022; Rycroft-Smith & Stylianides, 2022), and actionability – in terms of usefulness and direct support to application in practice (Cooper et al., 2017; Gleeson et al., 2024; See et al., 2016). Third, the research needs to be *integrative with other sources of knowledge*, as teachers heavily weigh research against their own professional expertise and the experiences and needs of members of the school community (Ahmed, 2019a; Baan et al., 2023; *Goodwin & Webb, 2014). The last theme is the *research's credibility*. Teachers often perceive educational research as contradictory (Cain, 2017) and disconnected from reality (Tunison, 2020b). Popular, well marketed formats (i.e. "How to" guides) and recommendations from colleagues increase trust (Graves & Moore, 2018).

The codes representing the arrow ($k = 70$) contain extracted data that addressed questions of why, where and when practitioners engage in research. The reasons and motivations for engaging in research described range from imposed use from leaders or as a part of requirements for academic coursework (Abbott et al., 2017; Mills et al., 2021; Penuel et al., 2017) to intrinsic motivations to provide the best learning for students or for practitioners own professional development (Abbott et al., 2017; Graves & Moore, 2018; Ion et al., 2024; Lysenko et al., 2015; Mehrani, 2015; Mills et al., 2021; Penuel et al., 2017). While the imposition of research seems rare, this very *lack* of imposition is framed both as a fundamental problem (Abbott et al., 2017) and as an essential prerequisite for practitioner agency (e.g. Riordan, 2022; Wescott, 2022; Yoshizawa, 2022).

In the included studies, colleagues and networks appear to be the most common source of research-based materials, although there are also other outlets sought by teachers and school leaders. Very often internet searches and unspecified internet sources are mentioned in the included studies (Abbott et al., 2017; Booher et al., 2020; Drill et al., 2013; Fraser et al., 2018; Hillmayr et al., 2023; Ilhan et al., 2015; Lysenko et al., 2014; Mehrani, 2015; Scheffler et al., 2024; Shi et al., 2022; Vanderlinde & van Braak, 2010; Williams & Coles, 2007). Another important source of research knowledge for practitioners is formal government outlets (Lysenko et al., 2015; Olmos-Rueda & Pattier, 2023; Williams & Coles, 2007) or information acquired in connection with formal professional development courses or conferences (Baildon & Ong, 2022; Ilhan et al., 2015; Mausethagen et al., 2018; Mills et al., 2021; Ratcliffe et al., 2005). However, the overwhelming consensus in the reviewed studies is that research presented through formal academic outlets is the least used source of research knowledge.

Finally, the extent and frequency of research use are represented by the "arrow" part of the initial framework, which illustrates how and under what conditions thoughtful engagement is applied with appropriate research. Interestingly, discussions of how often a

practitioner *should* engage with research are scarce. There are, however, distinct notions of it being *too seldom*, in the consistent and prevalent finding that teachers do not engage with research, illustrated by Ahmed and Pervin (2015) where the "overwhelming response" to whether teachers read research was "No. None of our participants had read research publications for their professional development" (Ahmed & Pervin, 2015, p. 28).

The final core component of research use is "thoughtful engagement and implementation" (Rickinson et al., 2022) ($k = 57$), and worth mentioning here, is that the coding has not taken into account the quality aspect of the QURE framework. Therefore, what is presented here are the modes and ways of engaging in research that are described in the included literature. And in this literature, there seems to be a wide consensus of what constitutes research (i.e. published, peer-reviewed, rigorous and systematic academic inquiry), and that "use" should entail engaging with this research to reach some ends that ultimately leads to better outcomes for students. In two series of included studies (Ahmed, 2016, 2019a, 2019b; Cain, 2015a, 2015b, 2017), the researchers delve deep into the discussions teachers have when they engage with (provided) research articles. In this specific corpus, it becomes very clear how teachers contest, adhere to, re-evaluate and interpret research findings and their own practice, in a way that lets the research serve as a "third voice" (Cain, 2015b) or that research use "supports neither wholesale rejection nor uncritical acceptance of research-based recommendations. Instead, it involves opportunistic decision-making." (Ahmed, 2019b, p. 724). When research is integrated with other sources of knowledge, and is engaged with in "long, focused discussions" (Cain, 2015b), it seems to serve as "thoughtful engagement" even though it is not always followed by "implementation".

Barriers and Enablers of Research Use

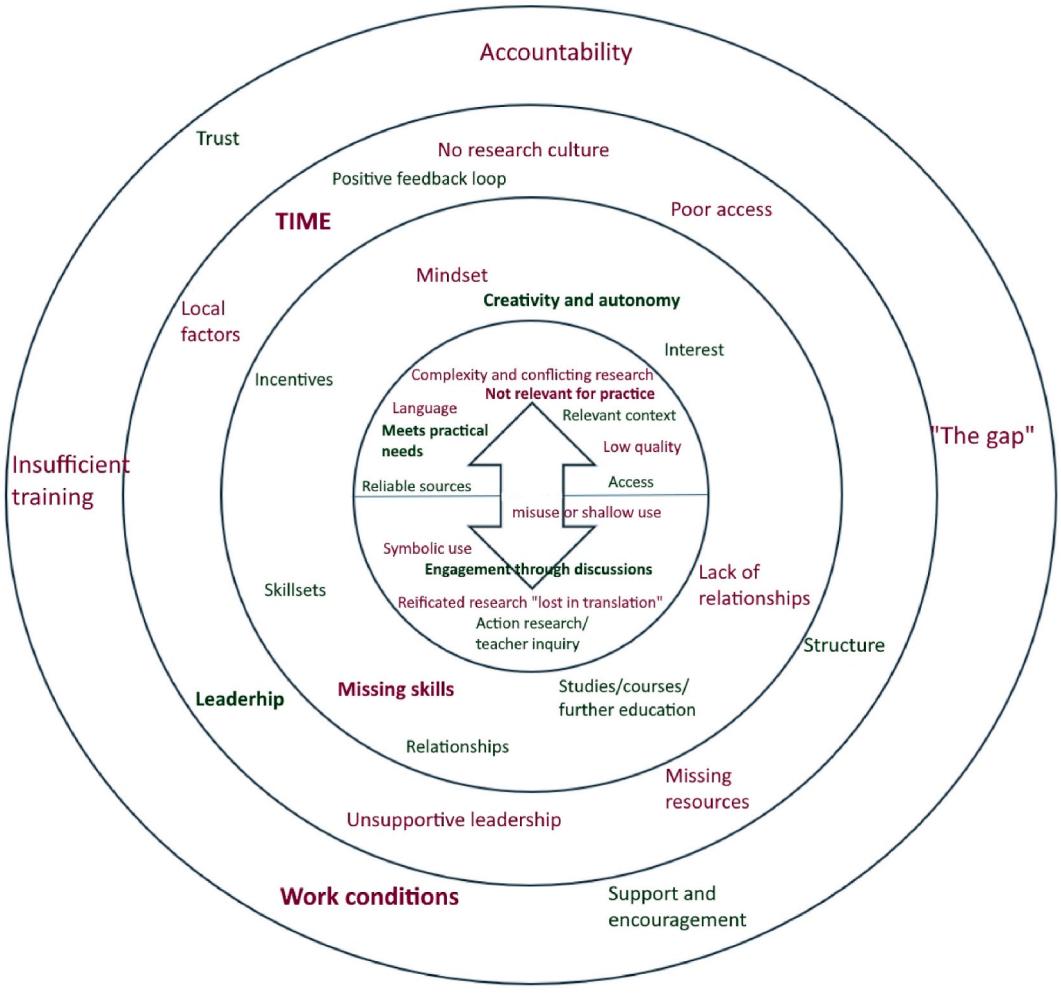


Fig. 3. Barriers and enablers of research use.

However, such long, focused discussions are rare, and more often in the literature the focus is of the lacking implementation (Davidson, 2013; Farley-Ripple, 2024; Kowalczuk-Walędziak & Ion, 2024; Levin et al., 2011), or the implementation of reified, symbolic or simplified interpretations or products (Déchène et al., 2024; Graves & Moore, 2018; Malin et al., 2019; Riordan, 2022; Tunison, 2020b; Wescott, 2022).

In the included studies we found examples of how engaging *in* research also may contribute to deeper understanding and meaningful use of research, in terms of practitioner inquiry or collaborations with researchers (Glaés-Coutts & Nilsson, 2021; Hartmann & Decristan, 2018; Sinnema et al., 2011). In these settings, however, the role of the researcher seems to be essential in leading to meaningful engagement, as illustrated in this quote:

> *That you don't feel you're coming out of university knowing something, have found something out and you're telling the pedagogical field about it, but instead (…) 'I'm coming along with my crazy ideas about supportive feedback – what does that look like?' and then I encounter the pedagogical field, and then I learn from them: This is great, what you did in your research, or your ideas on that, but that does not fit into our practice for this and that reason, because partly, you don't know what is going on here.* (Hartmann & Decristan, 2018, p. 17)

In sum, thoughtful engagement is a rarer phenomenon in our included studies, compared to instrumental and symbolic use of research. However, when researchers are very "hands on", and involved in the process of research use, they either seem to contribute to, or are able to observe deeper use, than when simply asking survey questions about the frequency of reading research or the sources practitioners seek.

3.4. Research question 2: barriers and enablers of research use in schools

Overall, the included studies show that the barriers and enablers of research use presented in the former research and frameworks still prevail. Fig. 3 shows how barriers (in red) and enablers (in green) were coded across different levels, with the codes in bold showing the most prevalent barriers and enablers from the included studies within each level.[1]

3.4.1. System level ($k = 36/k = 6$)

Barriers described on the system level include accountability ($k = 11$), insufficient teacher training ($k = 12$), school practitioners' work conditions ($k = 16$), and general factors pertaining to "the gap" between research and practice ($k = 13$). Accountability is described as a barrier in different ways and for different reasons. Some studies show that the results of a high stakes accountability system lead to a prioritization of material directly connected to tested curriculum (Scheffler et al., 2024; Sibanda & Begede, 2015), or that "High stakes exams are the priority in the use of time" (Baildon & Ong, 2022 p. 5). Using research also involves taking risks, as illustrated in this quote:

> Abe: So it's up to the teacher to flexibly incorporate and feel the gap?
>
> *Mary: Well, that is an interesting question because that is also what I ask myself a lot. You know, they give you very vague guideline; yet, I don't make the final exam. They make the final exam. So, another way of saying this is that you would decide what you want to teach, or how you want to teach, but we are not the ones who set the final exam. So, on the one hand, you have a lot of freedom to choose but by the same token, you better expose the kids what is on the exam. So, it is kind of interesting. The implication is clear. In the face of these state-mandated tests, teachers teach the students for the test, limiting both their readings and their instructional approaches to fit the requirements of the test. Using research or introducing research-based ideas under these circumstances is too much taxing for teachers. Simply stated, it means working against the system.* (Tekleselassie, 2006, p. 189)

Accountability as a barrier to research use is frequently mentioned, both as a threat to the freedom and autonomy of practitioners (Brown, 2017; Brown et al., 2022; Brown & Zhang, 2016) and as a "competition" to the legitimacy of research, perhaps due to a common conflation between data and research. As an example, Cain (2017) found that when he provided teachers with peer reviewed articles, teachers relied more on a report from the Office for Standards in Education (Ofsted), which details key performance ratings based on exam data and student outcomes.

Although accountability as a barrier most often seems to affect the low performing schools, in increasing risk-averseness and limiting autonomy, there is also an example of a high performing system where policymakers point to consistently good scores on PISA to *prevent* allocation of resources to schools, and to hinder teachers in changing their practices (Barrett & Prendergast, 2025). Accountability is also seen as a barrier in the production and dissemination of research both in terms of the "publish or perish" race (Vanderlinde & van Braak, 2010) and in the stepmotherly treatment of educational research, and the role of researchers in the education schools (Tekleselassie, 2006). These factors may also contribute to the perceived barriers of the gap between research and practice, which in some studies is referred to as a very common obstacle (Drill et al., 2013; Ion et al., 2021). This is exemplified by statements like "working in two separate worlds" (Hartmann & Decristan, 2018, p. 20), "a huge emptiness" (Davidson & Nowicki, 2012, p. 342) and *"The divide between spheres of research and practice in France is further complicated by incomprehension on both sides"* (Gaussel et al., 2021, p. 4).

Many studies identify inadequate training in research methods and activities within initial teacher education as a systemic barrier

[1] These numbers refer to the numbers of studies mentioning the barriers and enablers at the corresponding level, respectively.

to research use (Agbenyega et al., 2023; Ashur & Bagadood, 2022; Aukland, 2022; Barrett & Prendergast, 2025; Cooper et al., 2017; Gaussel et al., 2021; Hillmayr et al., 2023; Hosseini et al., 2024; Ion & Sirvent, 2022; Kowalczuk-Walędziak et al., 2024). Some also point to a lack of ability to develop research competence in continuous professional development initiatives (Cooper et al., 2017; Gaussel et al., 2021; Kowalczuk-Walędziak et al., 2024).

The work conditions of teachers are also mentioned as barriers to research use through extreme workloads (Aukland, 2022; Baildon & Ong, 2022; Boyne & Beadle, 2017; Brown, 2017; Cooper et al., 2017; Ion et al., 2024), low salaries and subsequently low morale (Ahmed & Pervin, 2015) and challenging classroom environments (Ashur & Bagadood, 2022; Davidson & Nowicki, 2012; Guckert et al., 2016). Teachers are preoccupied with other tasks, day-to- day "firefighting" and teaching in large classes comprising students with complex needs. In contrast to "time" as an organizational-level barrier, mainly referring to how schools allocate and structure teachers' working hours, the issue of "extreme workload" mentioned here operates as a broader system-level barrier affecting the workload of teachers and school leaders. Examples of this can be central labor agreements, accountability systems, and broader societal changes influencing the students.

Enablers on the system level are scarce in the included literature. One finding indicated that jurisdiction support can be a key enabler to research use (Prendergast & Rickinson, 2019) but more often suggestions are given to increase support via reform or legislation (Brown et al., 2022; Kowalczuk-Walędziak et al., 2024; Kowalczuk-Walędziak & Ion, 2024).

3.4.2. Organizational level ($k = 55/k = 39$)

The most mentioned barriers on the organizational level were "time" ($k = 46$) and "access" ($k = 31$). Together with resources ($k = 15$) and factors pertaining to students ($k = 2$), these constitute the structural organizational barriers found in the included literature. Conversely, many of the enablers in organizational structure were the opposite. Dedicated, scheduled and protected time is a critical enabler (Bergmark, 2022; Brown, 2017; Brown & Flood, 2018; Gleeson et al., 2024; Ion et al., 2021), and how this time is structured in formal meetings and facilitated collaboration is an additional enabling factor (Cornelissen et al., 2017; LaPointe-McEwan et al., 2017). Improved access through intermediaries and provided resources were also mentioned as structural enablers on the organizational level (Brown & Flood, 2018; Farley-Ripple, 2024; Hemsley-Brown, 2005; Vanderlinde & van Braak, 2010).

Organizational culture can also serve as both a barrier ($k = 18$) and an enabler ($k = 19$) of research use. In cultures that barrier research use, the engagement with research is scarce and something that is dealt with by individuals with a keen interest (Booher et al., 2020; Kowalczuk-Walędziak & Ion, 2024). The top-down implementation of policy is also connected to a lack of research culture, exemplified by a teacher who said that directives they are expected to follow are "plucked out of the air" (Boyne & Beadle, 2017 p. 19). Baildon et al. (2022) even suggest that the prevailing complaint of insufficient time serves as a proxy for the deeper issues of agency, support and a lack of voice in decision making.

In an enabling organizational culture, engagement with research can establish a positive feedback loop (Abbott et al., 2017) and many find that when teachers feel they work in an environment that values research, they are more likely to use research (Boyne & Beadle, 2017; Briand-Lamarche et al., 2016; Brown et al., 2016, 2022; Cornelissen et al., 2017; Gaussel et al., 2021). It is suggested that building a critical mass of practitioners who are committed to research use is a key strategy to enable a research culture (Hemsley-Brown, 2005). This also involves practitioners' positive beliefs, experiences and attitudes towards research use (Baildon & Ong, 2022; Davidson & Nowicki, 2012; Ion et al., 2021). Levin et al. (2011) also emphasized the importance of trust, and to foster a culture where open exchange of ideas and questioning of conventional wisdom is encouraged.

Key in this is leadership, which can serve as both a barrier ($k = 21$) and an enabler ($k = 25$) in the included studies. Leadership that encourages and advocates for research use enables it (Baildon & Ong, 2022; Brown & Zhang, 2017; Cooper et al., 2017; Levin et al., 2011). Another important enabling factor is building trust, autonomy and freedom (Baan et al., 2023; Brown et al., 2016; Kowalczuk-Walędziak & Ion, 2024). Furthermore, allocating time and cultivating collaborative cultures are also key in leadership enabling research use (Brown, 2017; Cornelissen et al., 2017; Drill et al., 2013; Kowalczuk-Walędziak et al., 2024). On the other hand, leadership can hinder research use by withholding support, or being indifferent to research use (Abbott et al., 2017; Ahmed & Pervin, 2015; Alhumidi & Uba, 2017) or by discouraging a research culture (Hosseini et al., 2024; Ion & Iucu, 2014; Kowalczuk-Walędziak & Ion, 2024). This can also occur as a result of other priorities by the school leadership, as illustrated by this quote from a teacher:

> *Leaders and Reporting Officers are not going to say that, 'Oh because you've done this research, it is well planned!' They just want it fast. I want this program, I want it by next week. [It] is more deadline-driven, more than research-driven, so, that doesn't contribute to the teachers' perceptions that research is important in schools.* (Baildon et al. 2022, p. 6)

In short, the findings from the included studies show that leaders who are described as proactive, supportive, and strategic can overcome the well-documented barriers. Conversely, leaders who are absent, indifferent, or obstructive effectively cement the barriers in place.

3.4.3. Individual level ($k = 44/k = 44$)

The enablers and barriers to research use described in the included studies can be found in individual skillsets ($k = 22$ in both), mindsets ($k = 16$ and $k = 29$, respectively), and in relationships ($k = 35$ and $k = 25$, respectively).

The individual skillsets that enable research use include both basic research skills (Ion et al., 2021; Lysenko et al., 2014) and the translational skills to apply research to context and practice (Kowalczuk-Walędziak et al., 2024; Lysenko et al., 2014). In addition, interpersonal and collaborative skills are highlighted as key enablers for participation in a research rich culture (Hemsley-Brown, 2005; Kowalczuk-Walędziak et al., 2024). Formal education and specialized training seem to be predictors of research use, (Davidson, 2013; Malin et al., 2019; Reichenberg, 2021), experience however, is not so clear. There are findings suggesting that

research use is most prevalent in early career (Akcoltekin et al., 2017) and in later stages of experience (Baan et al., 2023; Davidson & Nowicki, 2012).

Conversely, the lack of skills is prominently cast as one of the major barriers to research use. In our corpus we found both an inability to access and understand research (Ahmed, 2016; Akcoltekin et al., 2017; Tekleselassie, 2006), and the inability to critically evaluate research (Barrett & Prendergast, 2025; Booher et al., 2020; Levin et al., 2011). Practitioners also lack the confidence to apply research (Davidson & Nowicki, 2012; Prendergast & Rickinson, 2019) and fail to see the relevance of research (Baildon & Ong, 2022; Hudson et al., 2016). And as with the enabling factors, lack of skills is often traced back to inadequate training (Ahmed, 2016; Davidson & Nowicki, 2012; Gaussel et al., 2021).

The mindsets that enable research use can be tied to curiosity and professional ethics as intrinsic motivation. The first also involves creativity and courage, and fosters ownership and autonomy (Baan et al., 2023; Bergmark, 2022; Briand-Lamarche et al., 2016; Brown, 2017). Curiosity, or a deep personal interest in teaching and learning, appear linked to intrinsic motivation for research engagement (Hosseini et al., 2024; Kowalczuk-Walędziak et al., 2024), but nevertheless external incentives such as status, prizes and funding for academic coursework were also suggested as enablers in the literature (Abbott et al., 2017; Ashur & Bagadood, 2022; Kowalczuk-Walędziak & Ion, 2024). The commitment to professionalism, and the commitment to meet student needs, also serve as strong driving forces that enable research use (Kowalczuk-Walędziak et al., 2024; Kowalczuk-Walędziak & Ion, 2024).

The hindering mindsets described in the reviewed literature, can be summed up as a lack of trust, interest and perceived relevance of research, that in effect creates a fundamental disconnect between research and a teachers daily reality, and leads to apathy (Ahmed & Pervin, 2015; Baildon & Ong, 2022; Finnigan et al., 2013; Hwang, 2023; Sadeghi & Abutorabi, 2017). Fueling this, were fixed mindsets and comfort zone inertia (Ashur & Bagadood, 2022; Konstantinou, 2022).

The central theme of relationships enablers of research use in the included studies is connections, and conversely the central barrier is disconnection. The relationship enablers include both internal networks of trusted colleagues (Brown et al., 2016; Gleeson et al., 2024) and structured collaborations such as "journal clubs" (Boyne & Beadle, 2017), "enquiry groups" (Cornelissen et al., 2017) and communities of practice (Briand-Lamarche et al., 2016), where time and space for research discussions are created. Reaching outwards, both to other schools and districts (Hemsley-Brown, 2005; LaPointe-McEwan et al., 2017) and in partnerships with researchers and universities (Ion et al., 2021; Kowalczuk-Walędziak et al., 2024; LaPointe-McEwan et al., 2017) are also strong connections that foster research use.

However, when these relationships are lacking, the disconnect between researchers and teachers is perceived as operating in "completely separate" systems" (Hartmann & Decristan, 2018 p. 20) and there is a "huge gap" (Davidson & Nowicki, 2012 p. 342). The divide is characterized both by a lack of discourse (Déchène et al., 2024) and a mutual incomprehension of the other side (Gaussel et al., 2021). One reason might be that such collaborations consume and occupy more time than expected (Guckert et al., 2016). That might also explain how teachers also work in isolation within schools (Ashur & Bagadood, 2022; Briand-Lamarche et al., 2016).

3.4.4. Core components ($k = 55/k = 40$)

When looking at the core components described in the included studies, barriers and enablers can be found in both (thoughtful) engagement ($k = 5$ and $k = 18$, respectively) and in appropriate research ($k = 50$ and $k = 35$, respectively). Engagement can be "thoughtful" and serve as an enabler of research use both when it is instrumental, and engagement is driven by teachers to solve problems of practice (Barrett & Prendergast, 2025; Bergmark, 2022; Sinnema et al., 2011) or when its collaborative and leads to dialogues and discussions (Boyne & Beadle, 2017; Brown & Flood, 2018; Brown & Zhang, 2017). The engagement has also led to other

Revised Framework: Cross-cutting Multi-level Mechanisms for Research Use in Schools

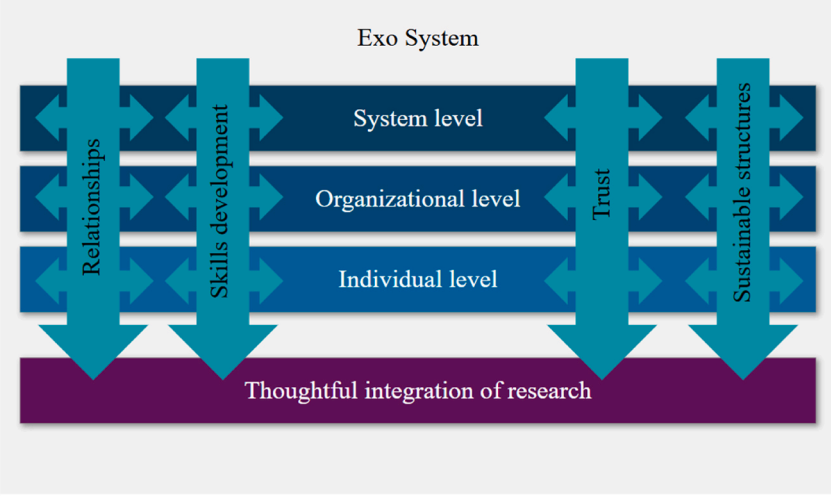


Fig. 4. Revised framework: Cross-cutting multi-level mechanisms for research use in schools.

tangible benefits such as informing concrete changes and increasing teacher agency, the latter well illustrated by this quote from Hofmann & Ilie (2022, p. 14) *"The good thing that I saw is getting staff engaged in research, seeing staff think, 'Oh, I can have some ownership in what I do', because we are a profession that are very much done unto(…)"*

On the other side, there are examples of superficial use of research, where it is used not to inform change, but to uphold status quo (Barrett & Prendergast, 2025; Davidson & Nowicki, 2012), in an instrumental, yet superficial way (Finnigan et al., 2013) or with such modifications that the practices were not recognizable from the original research (Guckert et al., 2016).

Whether "appropriate research" can serve as an enabler or a barrier to research use, seems to be contingent upon the appropriateness relative to different factors. Primarily the research needs to convey practical utility (Ahmed & Pervin, 2015; Baildon & Ong, 2022; Barrett & Prendergast, 2025; Lastrapes & Mooney, 2021), but there is a pitfall in oversimplification and distortion of the research (Booher et al., 2020; Sharples & Sheard, 2015). These factors apply to format, content, contextual relevance, and language. In order for research to be "appropriate", it needs to navigate the tensions in these factors both to serve the practice of the users, and to uphold the integrity of the research.

3.5. Configurative synthesis: four governing mechanisms of research use

Across the dataset, several themes emerged consistently across and beyond the individual, organizational, and system levels. These themes were evident both in the analysis of framings and descriptions of research use, as well as in the identification of barriers and enablers. Below, we propose that these recurring themes serve as four governing mechanisms shaping research use in schools: trust, skills development, sustainable structures, and relationships (see Fig. 4).

3.5.1. Trust

In the studies included, *trust* was a prevailing theme across levels. Outside the system level, trust applies to both the relationship between the general population and research, and also to the relationship that the population has with professional actors in education. The latter is often upheld by what can be called institutional distrust (Grimen, 2008), i.e., accountability systems that are set up to keep the profession within a restricted domain of agency, according to policy directions. The level of institutional distrust determines to which extent research use is a prerogative of professional judgement and agency. For example, a school system is likely to have aims that students learn to read at a relatively early age, and they are likely to have some form of assessment of what degree the professional actors within school deliver on this target. Whether or not the professional actors experience freedom to choose how young learners should learn to read, is a question of to which extent the institutional trust or distrust affect these choices: do the administrative part of the system impede a mandatory reading program, or are the teachers free to (ideally) discuss research and choose methods that are suited for their context and needs? This example shows how trust and distrust can govern research use from outside, and within the system level.

At the organizational level, trust is connected to culture, and the leadership that sustains a trusting culture. This entails a culture where trial and error, and innovations are supported and cheered on, rather than risk averse cultures that stay inert to change (e.g., Gaussel et al., 2021; *Greany & Brown, 2017). The connection to the mechanisms throughout the system level are clear here; a high-distrust environment becomes a tool for compliance and inertia through cultures "playing it safe", as opposed to high-trust environments that encourage innovation and professional judgement.

At the individual level, however, trust is not necessarily connected to the other levels as clearly, but a culture of trust seems to yield research engagement on the individual level, as seen for instance in Ion & Sirvent (2022), where teachers report that a trusting culture is fundamental to initiating experiments in their classrooms. More importantly, the individual trust is connected to the exosystems societal factors of general trust in research. Although actors within schools have a slightly higher trust in research in general than the population at large (Rosman & Merk, 2021), the mechanisms that drive skepticism, trust and distrust in the society also affect teachers and school leaders. Epistemic trust - trust in the sources of research, from the recommendations from colleagues to the trust in renowned researchers and outlets, are also mechanisms that govern individuals' use of research.

3.5.2. Skills development

The second theme that was prevalent in the corpus of included studies, and that spanned across the levels, was *skills development*. At the outside of the system (the exosystem) there are political directions on how education and teacher education should serve their purposes to society. Longstanding discussions on the relationship between theory and practice in teacher training and higher education belong outside the school system but affect and govern research use within it. In our included studies, we find that teachers and school leaders that have some research experience from their initial education tend to engage more in research than those who do not (Kowalczuk-Walędziak & Ion, 2024; Ratcliffe et al., 2005; Williams & Coles, 2007). However, the skills required to utilize research effectively also apply to policy and leadership at the system level. If these skills are lacking at the top, it is often reflected in fragmented research use in the levels below. Symbolic and shallow use at the system level is often exposed as such by actors at the levels below, as seen in Wescott (2022) and Boyne & Beadle (2017). And the same could be observed about skills development at the organizational level, where there both seems to be connections between skillful leaders and the development of research engaged culture on the one hand, and strategic, symbolic and shallow grappling with research on the other.

As thoroughly handled in the included studies, the individual skills to search for, find, assess and integrate research, are also mechanisms that seem to be related to how, and to what degree, research is used by individual teachers and school leaders in schools. The conditions needed for the development of skills, especially at the organizational and individual level, are also related to the third theme we argue as a governing mechanism of research use: sustainable structures.

3.5.3. Sustainable structures

Sustainable structures are the mechanisms that govern the allocation of time and the working conditions that are crucial to promoting research use in schools. In the exosystem this involves structuring of the school systems in terms of teachers and school leaders' work agreements, students' curriculum and timetables, and the general structures that affect the working conditions of actors within school systems. At the system level, this can involve funding and initiatives to support professional development, as well as structures that bridge different institutions within and outside the system – which we further introduce in the final theme. At the organizational level, the sustainable structures are again connected to leadership, first and foremost in the structuring of time and spaces for research engagement (e.g., Friesen & Brown, 2024; *Greany & Brown, 2017; Lysenko et al., 2015). The sustainability aspect is added to showcase the tension that can be found between societal needs for education, the economic costs of education, and the "environment" of the professional values within school systems. This structuring needs to adhere to all three in order to be sustainable. Sustainability, in this context, refers to the shift from research use as a temporary project to its status as an integrated, long-term feature of the professional landscape.

At the individual level, structural conditioning needs to result in the professional responsibility to take advantage of the time and space allocated for engagement with research. Although this was not extensively described in our included studies, there were some examples of teachers doing something else, during time allocated for research discussions by the researchers who studied it (See et al., 2016; Williams & Coles, 2007). Mostly this was explained by the researcher as a lack of interest or ability to engage with research. But again, structural factors in relation to teachers' work conditions at the higher levels can directly be related to individual ability to use research. In poor working conditions and a stressful environment, teachers are more likely to burn out than to pick up a research paper.

3.5.4. Relationships

The final prevalent theme from our findings, that we see as a governing mechanism across levels of research use, was *relationships*. Starting in the exosystem we can see the infamous "gap" between researchers and practitioners as a factor that can condition research use in the nested levels beneath (Davidson & Nowicki, 2012; Gaussel et al., 2021; Hartmann & Decristan, 2018). This is perhaps the clearest mechanism that can be found governing research use at all levels. Narrowing of, or closing the gap, involves mutual relationships between researchers, policy makers at the system level and practitioners at the organizational and individual level. These relationships can take the form of formal courses and professional development at different levels, or in research-practice-partnerships. The key seems to be some degree of mutuality in terms of respect for each other's jurisdiction of expertise, and open dialogues based on trust and curiosity. Some of our included studies (e.g., Cain, 2015b; Cornelissen et al., 2017) showed this relationship clearly at the individual level, both when the researchers were part and parcel of the research use under study, and when the practitioners themselves engaged in research through action research or professional inquiry.

Relationships between systems, schools, and individuals are also driving forces of research integration in practice and are connected to the structures described in the previous section. In sum, we argue that relationships cannot be tied to individual factors that pertain to research use but are rather one of the main mechanisms that govern research use in schools, across levels.

4. Discussion

This systematic review investigated how research use is framed and described in the extant empirical research on research use in schools, and identified what barriers and enablers of such use can be found in the literature. It integrated multiple strands of evidence to advance a revised, multi-level framework that explains how four governing mechanisms shape research use in education. It has also demonstrated that use of research in schools is a multi-faceted phenomenon, shaped by a complex interplay between individual, organizational and systemic factors as well as factors at the periphery of the system. As a theoretical lens, we started out with a consolidated framework of recent conceptualizations of research use, and this enabled us to see some underlying mechanisms that govern research use in schools at multiple interconnected levels. Across the literature, a clear distinction was made between a linear and instrumental approach to research use, and a model centered on professional agency and collective sense-making. The latter entails practitioners' thoughtful engagement with and integration of research, considering context, practicality, credibility, and integration with other sources of knowledge. Four mechanisms appear to govern such use across, and from outside the school system: trust, skills development, sustainable structures, and relationships.

These four mechanisms can be seen as mutually reinforcing and operate simultaneously across levels in both vertical and horizontal directions. Trust is a precondition for the kinds of open dialogue and risk-taking that skills development and relationship-building require but can also be sustained by and through them. Sustainable structures create the time and space within which relationships can form, and skills can grow. Relationships, in turn, generate social infrastructure through which trust is built and sustained, and through which structural resources are used effectively rather than symbolically. Skills development both rests on and reinforces all three other mechanisms; it requires trusting environments and collaborative relationships to flourish and is most durable when embedded in sustainable structures. In this sense, the four mechanisms form an integrated system rather than a checklist, and their absence or presence at any level reverberates through the others.

This configurative synthesis also marks the key point of departure between the initial consolidated framework in Fig. 1 and the revised framework in Fig. 4. The initial framework, built around the QURE model and supplemented by concepts from Eriksen (2022), Godfrey & Brown (2019), Farley-Ripple et al. (2018), Kvernbekk (2016) and Perry and Morris (2023), organized research use into nested ecological levels with components assigned to each. The empirical findings from the included studies broadly supported the relevance of these components, and the level structure has been retained. However, the aggregative analysis indicated that the most persistent barriers and the most enabling conditions could not be adequately explained by level-specific factors alone. What the

findings from included studies suggested was that certain conditions seem to govern research use regardless of level.

Based on our findings, the revised framework therefore replaced the nested component logic with four cross-cutting mechanisms. The findings suggest that, firstly, the most persistent barriers and enabling conditions operate across levels rather than within them; secondly, these conditions interact dynamically rather than independently; and thirdly, understanding research use requires a framework that captures this cross-level interaction rather than assigning factors to discrete levels. Although not explicitly incorporated in the framework, leadership can be understood as a catalyst for these mechanisms, while culture may be viewed as one of their outcomes.

As can be seen through this analysis, we have shifted from "thoughtful engagement and implementation" of research, to "thoughtful engagement and *integration of research*" in our narrative. This distinction relates to but extends beyond conceptual use. Conceptual use in the Weiss' (1979, 1980) tradition describes how research gradually shifts thinking without a direct application, whereas "integration" implies a more active and deliberate process of reconciling research with professional wisdom and contextual knowledge.

Integration here can be understood in two complementary senses. Epistemologically, it refers to the active reconciliation of research knowledge with professional wisdom, contextual judgment and the experiential knowledge of teachers and school leaders - research as a third voice in professional dialogue rather than a directive. Organizationally, it refers to the embedding of research engagement into the culture, routines, and structures of schools as a sustained collective practice rather than an individual or occasional activity. These two senses can seem symbiotic: epistemological integration requires organizational conditions that make sustained engagement possible, while organizational embedding is only meaningful if it supports genuine rather than symbolic epistemic engagement.

This shift was grounded in a prevalent finding across the included studies: research consistently plays a supporting rather than leading role in practitioners' decision making and improvement, remaining a 'third voice' in a three-way dialogue between research knowledge, professional experience and contextual wisdom. Moving from implementation to integration therefore implies that it is at the professional's discretion to integrate research where it is fit for context, practice and the needs of school leaders, teachers and students, rather than imposed from levels above. This aligns with trends in the knowledge mobilization literature, where conceptualizations have moved from linear models of knowledge transfer toward relational and system models in which knowledge production and use are dynamically connected (Torres, 2022), and where integration has emerged as a key concept describing the active embedding of research into practice through ongoing dialogue between researchers and practitioners rather than through dissemination alone (Lockton et al., 2022).

How this integration operates in practice can be traced through each level of the revised framework. Our initial framework was built around the QURE framework (Rickinson et al., 2022), using concepts and elements from other frameworks as sub-themes or explanatory constructs. In the QURE framework, the system level is not specified beyond being a complex source of influences. Thus, we additionally used the framework of Godfrey & Brown (2019) to get a deeper understanding of the components of research use at that systemic level. In their ecosystem, the system level is the macrosystem and the exosystem where we can find (intelligent) accountability, trust and societal aims and expectations for education, as well as a "belief in key role for practitioners and researchers" and a "strong inclusive narrative about importance of research informed practice" (Godfrey & Brown, 2019, p. 212). These elements align with our proposed mechanisms that start in the exosystem and affect the nested levels beneath. However, the empirical findings are scarce at this level, showing us that research on research use to a greater extent has focused on individual professionals in schools. A potential consequence of this is that we continue to find the same barriers to research use, because accountability systems and institutional distrust as part of the narrative about the importance of research informed practice, continues to translate into pressure to implement standardized evidence-based practices.

The organizational level of QURE consists of infrastructure, culture, and leadership, and we initially specified that infrastructures include support systems and knowledge mobilization networks. In this study, knowledge mobilization networks were conceptualized as operating across organizational and system levels rather than being located exclusively within the organization. Although Godfrey & Brown (2019) represent knowledge mobilization as a bi-directional arrow linking and mutually reinforcing a research-engaged school with the wider system, we interpret this framing as consistent with our broader emphasis on relationships that span and connect multiple levels rather than residing within a single level. This interpretation is more aligned with our shift towards understanding processes as cross-level and relational rather than strictly nested.

At the organizational level, culture emerges from our findings as adaptive and semi-permeable. The latter corroborates Kvernbekk's (2016) notions of system flexibility and stability. A semi-permeable culture needs enough stability to maintain core values but enough flexibility to let research "disrupt" status quo thinking. From our findings at the organizational level, there seems to be a strong feedback loop, or positive spiral that can allow this to happen. The three components are interdependent and mutually reinforcing each other, with leadership as the main catalyst by allocating resources and creating a vision. Structure provides the time, roles, and routines to operationalize that vision. Culture represents the social and psychological environment that determines whether these structures are used effectively and sustainably. In summary, when these components are aligned, research use is framed as an embedded, collective, and reflective practice. When they are misaligned or absent, research use remains a sporadic, individual, and often inefficient activity.

In the included studies, the individual level of both mindsets and skillsets were widely described, but with scarce discussion on how these were connected to mechanisms higher up in the system. We argue that skills development is a wider responsibility, that needs to start with a discussion on how both initial teacher training and ongoing professional development can serve as enablers for individuals' thoughtful engagement and integration of research in their practice. In addition, discussions on what professional agency entails regarding research use need to happen with the sensitivities of research literacy in mind (Eriksen, 2022). To uphold research literacy as

a professional virtue, knowledge from research needs to be reconciled with practical wisdom.

At the core of the revised framework, the questions practitioners ask when engaging with appropriate research, concerning context, practicality, credibility, and integrity of other knowledge sources, align closely with Kvernbekk's (2016) conditions for permissible application of research findings.

Across all levels, the findings converge on a consistent picture: research use in schools is not primarily constrained by individual deficits but by the systemic conditions that govern whether trust, skills, relationships and structures can develop and interact productively. Taken together with Mausethagen et al.'s (2025) finding that deficit framing has dominated conceptualizations of research use, the governing mechanisms proposed here suggest that this deficit is not primarily located in individual teachers but is systematically produced by the interaction of trust, skills development, sustainable structures and relationships across levels. a finding that reinforces their call for more profession-sensitive approaches.

4.1. Implications

The implication of this systematic review is that there needs to be a shift in how we understand and study research use, moving from looking at the individual skills and attitudes of professional actors within the system, to the systemic mechanisms that govern research use, within and outside the system. Use of research cannot be mandated and imposed through policy directives, but should rather be supported through trust, skills development, relationships and sustainable structures that support the integration of research into professional practice.

The activities of research use should also shift towards this integration – searching, interpretation and participation in research use – should then be seen in the light of the context and practicality of the research users on all levels. With research integration, thoughtful engagement involves deep and critical reading (or other ways of consuming research materials), followed by long, focused discussions with colleagues and/or researchers. The output may take the form of a short-term change in thinking, an exploration of potential changes in practice, or a seed of thought that could lead to longer-term change.

We therefore proposed a revised multi-level framework of the mechanisms that can govern research use, that can function both as a conceptual lens on further research on URE, as well as a guide for policymakers and practitioners who wish to enhance their use of research when improving practice. The revised framework should be tested in future research.

4.2. Limitations

During the preparation of this systematic review, it became evident that the high volume of search results required a narrowing of scope. Consequently, theoretical and conceptual papers were excluded, as well as empirical research not published in peer-reviewed journals or in other languages than English. While the search strategy was systematic and conducted across three major databases with a pre-registered protocol, it did not include hand-searching of reference lists or grey literature sources such as reports and dissertations, hence studies meeting the eligibility criteria that were not indexed in the searched databases may therefore have been missed. This resulted in a restricted publication window that may have failed to capture historical shifts in how research use has been framed over time and across cultures. However, this was mitigated by the initial framework, which synthesized established theoretical and conceptual work to ensure earlier perspectives were represented.

The revised framework was developed primarily from studies conducted in Anglophone and Western European educational contexts, which tend to share certain structural features, including relatively high degrees of school autonomy, established traditions of professional development, and functioning research infrastructure. The applicability of the four governing mechanisms in centralized systems, post-socialist educational contexts, or low-resource settings remains an open empirical question. In highly centralized systems, for instance, the mechanism of trust may operate very differently if institutional distrust is structurally embedded in top-down governance. In low-resource settings, sustainable structures may face constraints that render the framework's recommendations aspirational rather than immediately actionable. These contextual variations represent productive directions for future empirical work that could refine and extend the mechanisms proposed here.

Furthermore, while this study proposes a revised multi-level framework for understanding the mechanisms governing research use, it does not make causal claims regarding the relationship between different operationalizations of research use and their outcomes. Further research is necessary to establish whether such causal connections exist.

The variation in quality of our included studies was taken into account in the analysis, but the evidence from the studies of low quality does not affect our overall findings, as the findings across several high quality studies were consistent with our revised framework.

5. Conclusion

By integrating different frameworks, and aligning them more closely with empirical findings, this systematic review offers a coherent understanding of how systemic factors may shape the internal capacity of schools to transform research use from a symbolic requirement into a potential catalyst for educational change. Ultimately, achieving meaningful impact requires a fundamental shift in orientation: moving from the mere adoption of pre-packaged findings to the long-term sustained integration of research as a 'third voice' in professional dialogue and whole-school culture.

Author contribution

Camilla Gramstad Hagevold: *Project administration, Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review and editing.*

Sanna Erika Forsström: *Conceptualization, Methodology, Software, Validation, Investigation, Writing – review and editing.*

Serap Keles: *Conceptualization, Methodology, Software, Validation, Investigation, Writing – review and editing.*

Ethical statement

This review does not require ethical approval.

Declaration of generative AI and AI-assisted technologies in the writing process

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Supplementary data

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Data availability

Data will be made available on request.

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